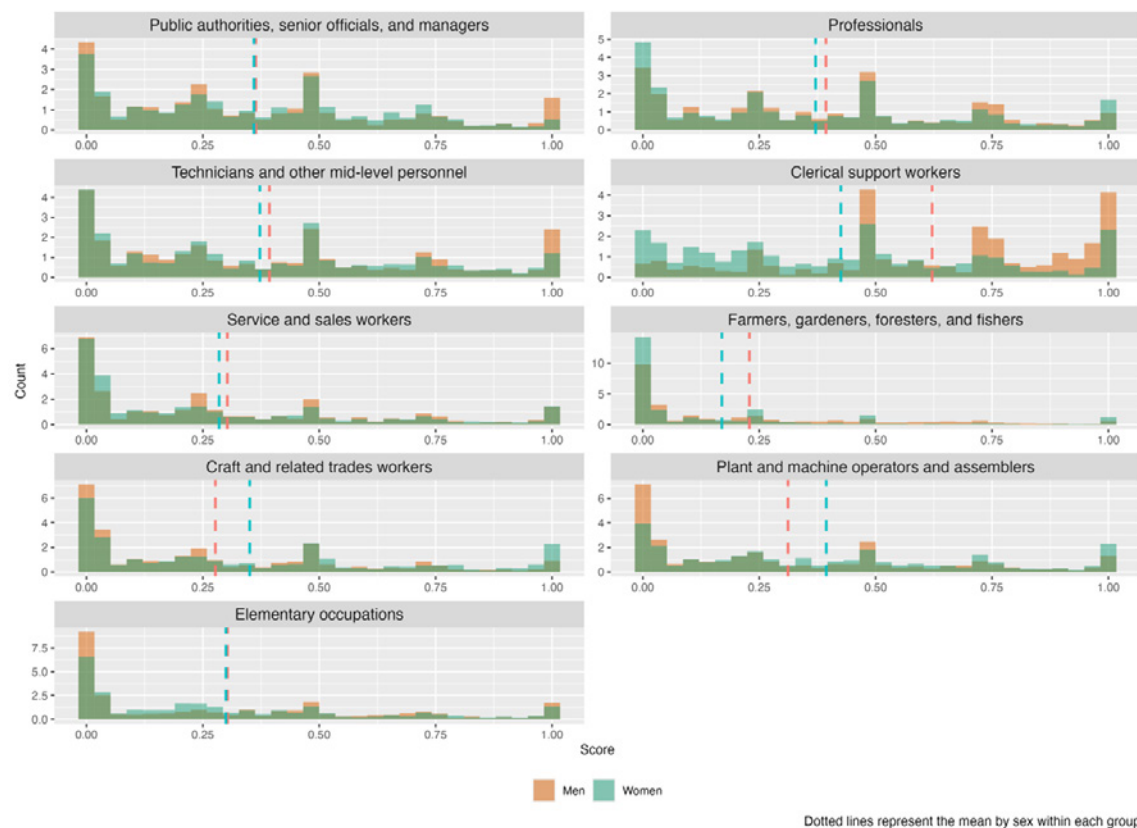


reflecting a different perception of task automatability or biases regarding clerical work, despite their lower representation in this group. Conversely, men dominate responses in occupations such as craft and trade workers and plant and machine operators, but for the tasks in these occupations, they assign lower scores than women. These differences may be representative of different task composition or even tasks performed by men and women, even when in the same occupational category, which then influence their scoring.

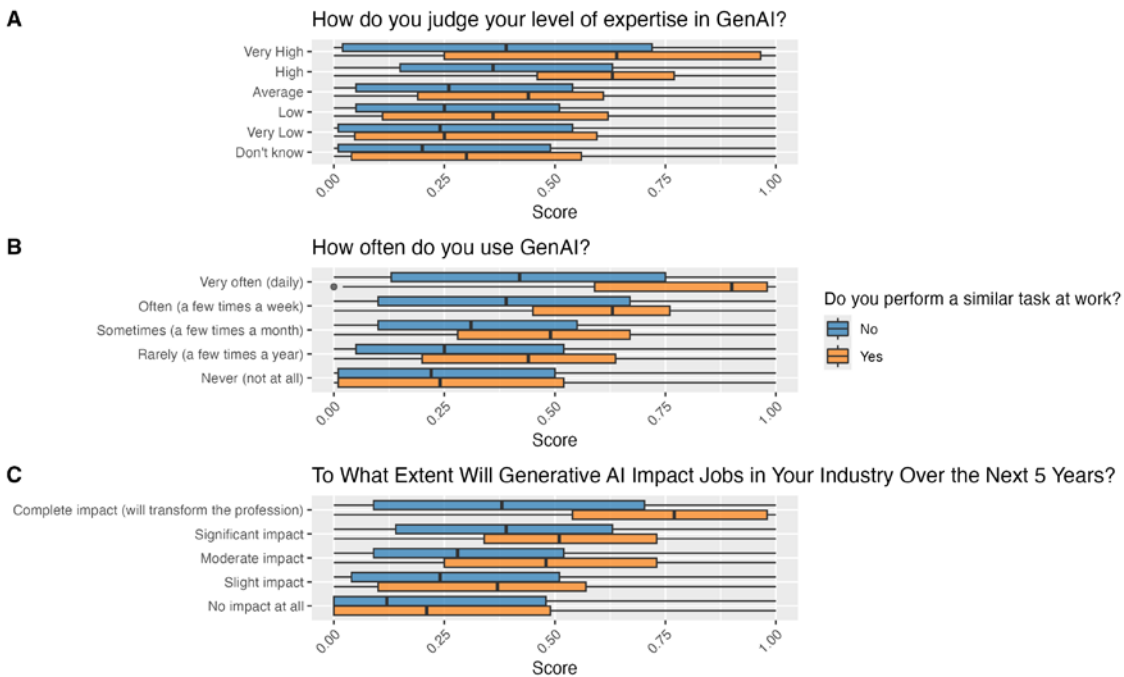
► **Figure 9. Distribution of task-level scores by sex and occupational group (1-digit ISCO-08)**



To better understand the factors associated with higher task automation scores, we conducted a set of exploratory regressions using the main task-level score as the dependent variable and controlling for occupational clustering. Results suggest that across occupations, familiarity with a task is associated with higher scores, and this relationship strengthens with greater reported exposure to GenAI. Respondents who use GenAI frequently – such as on a daily or weekly basis – tend to assign higher scores. Similarly, those who believe GenAI will have a “Complete” or “Significant Impact” on their occupation consistently assign the highest task scores (Figure 10).⁸

⁸ These results are based on exploratory regressions conducted for descriptive purposes. Full model specifications, diagnostics, and robustness checks were beyond the scope of this paper, as the survey serves as a starting point in a multi-layered refinement method. A more comprehensive econometric analysis of scoring behavior, including model selection and treatment of potential confounders, could be the focus of a separate methodological paper.

► **Figure 10. Scoring as a function of familiarity with GenAI and the scored task – distribution of individual scores**



One way to address this variation in scoring would be to apply corrective weights to responses, giving greater influence to respondent profiles likely to have more relevant expertise. For example, one could assign more weight to respondents reporting greater familiarity with GenAI or frequent usage or prioritize scores from individuals who perform similar tasks in their daily work. However, in our sample, this would introduce significant limitations. For 91.5% of the tasks, respondents did not report performing a similar task in their work – even when the task belonged to the same 1-digit occupational group as their current job. This is not surprising. For instance, within the “Professionals” group alone, the representative sample consisted of 368 tasks drawn from a pool of 8,715 tasks (Table 2), making exact task matching unlikely.

Moreover, adjusting for beliefs or usage would require strong theoretical assumptions about what constitutes “credible” scoring. For example, a respondent who believes their profession will be completely transformed by GenAI in the next five years may be highly informed, or may be expressing anxiety, speculation, or dystopian fears. Without richer data to distinguish between these interpretations, applying corrective weights risks introducing new biases.

Given the relatively small size of these specific subgroups in our dataset, we refrain from implementing such adjustments in the current analysis to avoid overfitting or compromising credibility. Instead, we flag this as an important area for future survey design, particularly in capturing respondent expertise and task familiarity more systematically.

For the present study, we treat the survey as a starting point, designed to capture a broad spectrum of perspectives reflecting varying degrees of familiarity with both tasks and GenAI. In many cases, a respondent’s understanding of the task itself may be more influential than their understanding of the technology, especially for concrete physical tasks (e.g. trimming flowers, assembling mechanical parts). Accordingly, we rely on mean scores across all respondents to reflect the collective judgment of a diverse sample grounded in workplace experience. To enhance

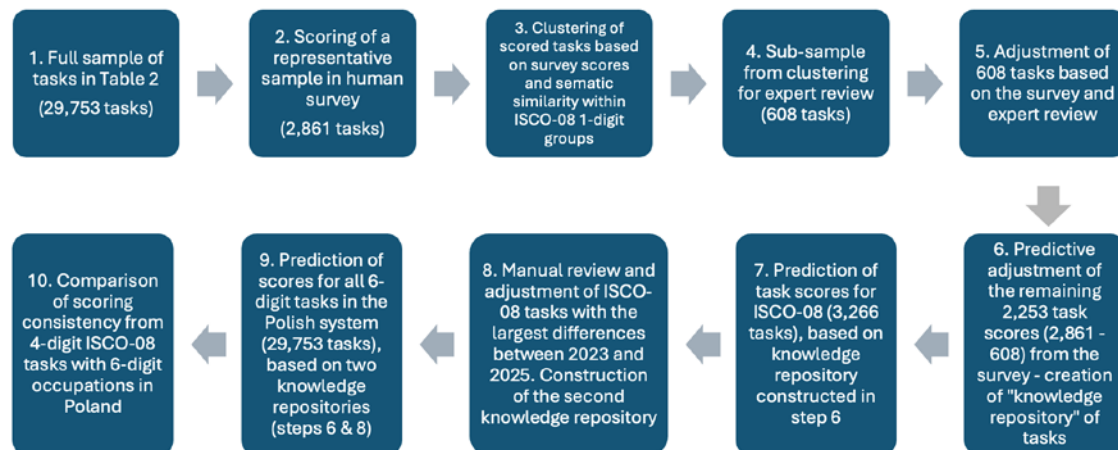
robustness, we supplement this with expert evaluation of a subset of tasks (section 5), ensuring that final scores reflect both practical insights and a realistic appraisal of current GenAI capabilities.

► 4 Expert validation survey

The expert group includes international specialists at the ILO and its partner organizations, NASK and the Polish Ministry of Family, Labour and Social Policy.⁹ The primary criterion for expert selection is their understanding of labour markets, including the complexity of occupations and tasks, and their real-life applications. While the experts are expected to have familiarity with the current family of generative AI technologies and their applications, expertise in AI design or development is not a requirement. This approach ensures that the focus remains on the feasibility and practical implications of automation within existing technological constraints, as technology experts tend to overstate technological capacities and risks when evaluating broader applications, particularly in workplace settings (Karger et al., 2023). This tendency is evident in earlier automation studies, which often assigned high displacement potential to routine tasks or entire occupations, such as garment production. As de Mattos et al. (2020) illustrate, these technological predictions frequently failed to account for the challenges of automating tasks requiring high dexterity or handling complex materials, such as stitching pliable fabrics. Accordingly, the assessment of automation potential in our study prioritizes expertise of task execution and workplace realities over purely technological expertise.

Since the experts cannot review all 2,861 tasks used in the survey, we focus on a representative sub-sample, selected in such a way to subsequently permit adjustments to entire task clusters. The overall scheme of this score adjustment procedure is presented in Figure 11.

► Figure 11. Schema of the scoring stages



First, we cluster the tasks into four fixed intervals, based on the distribution of the mean scores from the survey.¹⁰ We then use hierarchical clustering to group similar tasks within those clusters based on their semantic content. First, we assign embeddings to each task description using the *text-embedding-ada-002* model from OpenAI. Next, for each subgroup – defined by the combination of an occupational group at ISCO-08 1-digit level and a particular survey score cluster – we

⁹ A full list of contributors is provided in the acknowledgements section.

¹⁰ $\mu \in [0, 0.1] \cup (0.1, 0.3] \cup (0.3, 0.5] \cup (0.5, 1]$

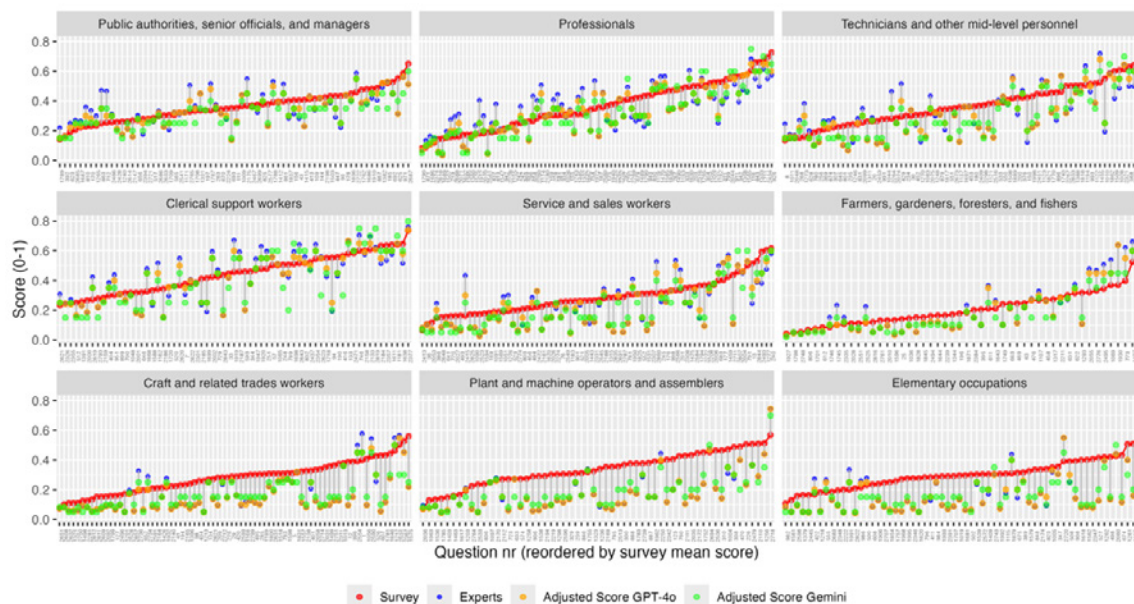
Each of the selected 608 tasks was then presented to a panel of experts. For each task, the experts were shown (i) the occupation and cluster information, (ii) the average survey score (rescaled to 0–1), and (iii) the full task description. If they agreed with the survey score, they left it unchanged; otherwise, they entered a revised expert score (0–1).

In many cases, the differences between the survey and expert scores are non-trivial. They also vary significantly across occupational groups, revealing distinct patterns based on the nature of pertaining tasks (Figure 13). In occupations dominated by manual and physical tasks, such as Craft and Related Trades Workers, Plant and Machine Operators, and Farmers, Gardeners, Foresters, and Fishers, experts consistently assigned lower scores compared to the survey respondents, reflecting the practical limitations of automating tasks that require physical manipulation or manual dexterity. In contrast, occupations involving more cognitive or knowledge-intensive tasks, such as Clerical Support Workers and Professionals, show a much more mixed picture. While some tasks in these groups received lower scores, many others were revised upwards by the expert group, reflecting the greater adaptability of generative AI for structured, data-driven, or communication-focused tasks. For example, Clerical Support Workers display a combination of increases and decreases, suggesting nuanced task-specific assessments where automation potential depends heavily on the nature of individual tasks.

► 5 Adjustment of survey sub-sample

In the next step, we compared the average of expert scores with the original mean survey scores using two independent AI models as impartial arbiters to propose an adjusted score.¹³ The models were informed about the sources of the scores in such a way that would prevent them from automatically siding with the expert group: group one was defined as *“practical knowledge of the work and types of tasks they were presented, since they worked in the same occupational groups from which we selected tasks they scored”*, while the second group was described as *“respondents with more theoretical knowledge of technology and labour markets”*. Rather than simply averaging the two scores, the models were explicitly instructed to focus their analysis on the task characteristics and the current capabilities of generative AI and required to provide justifications for each adjustment and its relative position to the original scores.

► Figure 13. Task-level scores from the survey and experts, compared to AI-arbitrated scores



The overall correlation between the adjusted scores produced by the two AI models is 0.96, with correlations within occupational groups exceeding 0.9 in all but one case¹⁴ (Figure 14). This high level of agreement demonstrates that, when provided with sufficient background knowledge in the form of reference ranges of scores, the models converge on remarkably similar predictions. Notably, only three tasks exhibited an absolute difference in scores exceeding 0.2 (orange dots in Figure 13). In these cases, GPT-4o tended to adopt a more optimistic perspective on the feasibility of automation, whereas Gemini emphasized the importance of human judgment as a critical factor.

¹³ We used OpenAI's GPT-4o and Google's Gemini Flash 1.5, which have comparable abilities. We also tested OpenAI o1, which at the time of writing was among the models with strongest capacities of « algorithmic reasoning » and observed very similar results. The process was set up as a cycle of sequential API calls, which loop over each task and capture the response into a data frame.

¹⁴ Public authorities, senior officials and managers in Figure 10.